



Phase Change Material Heat Exchanger Experimental Analysis

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Summary

The project aims to enhance Heat Exchanger (HEX) efficiency by optimizing fin impact through CFD analysis, including assessing Phase Change Material (PCM) HEX performance under varying inlet velocity conditions, including scenarios involving natural convection only. The overarching objective is to balance flexible airflow with optimized heat transfer rates and overall efficiency, while also gauging the suitability of PCM HEX for latent heat storage (LHS) applications within dwelling spaces.

To achieve these aims, the project has set specific objectives. Firstly, it will evaluate the impact of incorporating fins into the HEX to enhance heat transfer efficiency, comparing HEX designs with and without fins through CFD simulations. Then, it will optimize fin spacing within the internal air duct to ensure optimal heat transfer performance. Additionally, the project will investigate the influence of inlet velocity on HEX performance, assessing thermal energy transfer and discharge times across various velocities. Finally, it will assess the suitability of the HEX for natural convection airflow scenarios, aiming to determine the feasibility of relying solely on natural convection for airflow through the duct.

The CFD analysis revealed that fins significantly enhance heat transfer within the duct, particularly with an optimal fin spacing of 8mm, improving both thermal efficiency and reducing pressure drop by half compared to a 5mm spacing. This spacing optimization not only enhances performance but also enables the use of lighter, less costly fans. Mathematical validation ensured accuracy in the analysis. Additionally, the HEX demonstrated effectiveness across various velocities, enabling adaptable discharge rates for PCM, suitable for inconsistent heating needs in dwelling spaces. Moreover, natural convection alone sustains HEX operation for 6 to 10 hours, further reducing reliance on fans. These findings, combined with previous optimizations, position the HEX as an efficient, cost-effective solution for thermal energy storage in residential settings.

Moving forward, future work involves constructing a detailed CFD model specifically for simulating natural convection within the PCM HEX. This advanced model aims to provide more accurate insights into the behaviour of the HEX under natural convection conditions within the duct. However, constructing such a model poses challenges, including surpassing ANSYS student's one million node limit and navigating increased complexity during setup. Nevertheless, overcoming these hurdles promises a more detailed and reliable analysis, crucial for determining the viability of removing the fan from the HEX in residential applications.

Contents

NOMENCLATURE.....	4
ACRONYMS	4
1. INTRODUCTION.....	5
1.1. Background and Context.....	5
1.2. Aim and Objectives	7
2. LITERATURE REVIEW	8
3. METHODOLOGY	10
3.1. Methodology of Experiments.....	10
Analysis of the Effectiveness of Fins	10
Optimisation of Fins Analysis.....	10
Effect of Velocity Analysis.....	11
3.2. CFD Model Setup.....	12
Analysis of the Effectiveness of Fins	12
Optimisation of Fins Analysis.....	13
Effect of Velocity Analysis.....	13
4. Validation	14
4.1. Optimisation of Fins Analysis	14
5. RESULTS AND DISCUSSION	16
5.1. Effectiveness of Fins Analysis.....	16
5.2. Optimisation of Fins Analysis	18
5.3. Effect of Velocity Analysis	22
Calculations	22
Variations of Inlet Velocity	24
Natural convection	26
6. CONCLUSIONS	27
7. FUTURE WORK.....	27
8. REFERENCES	28
9. APPENDIX.....	30
Optimisation of Fins – Refined Test Temperature Profile	30

NOMENCLATURE

Symbol	Quantity	Unit
T	Temperature	$^{\circ}C$
x	Distance	m
h	Convective Heat Transfer Coefficient	$\frac{W}{m^2 K}$
P	Perimeter	m
k	Thermal Conductivity	$\frac{W}{m K}$
A	Cross-sectional Area	m^2
C_p	Specific Heat Capacity	$\frac{kJ}{kg K}$
ρ	Density	$\frac{kg}{m^3}$
Q	Thermal Energy	kWh
$\dot{Q}$	Thermal Energy Transfer	kW
$\dot{m}$	Mass Flow Rate	$\frac{kg}{s}$

Subscripts/ superscripts

i	inlet
o	outlet
∞	Surrounding air

ACRONYMS

Acronym

PCM
HEX
LHS
CFD
CAD
TES
CPU

Meaning

Phase Change Material
Heat Exchanger
Latent Heat Storage
Computational Fluid Dynamics
Computer Aided Design
Thermal Energy Storage
Central Processing Unit

1. INTRODUCTION.

1.1. Background and Context

This project aims to improve on the phase change material heat exchanger (PCM HEX) that has already been made. The PCM HEX holds 5kWh of thermal energy, but some areas require further analysis and optimisation to make the most efficient use of this. One example is whether the fins within the duct are necessary for the function of this PCM HEX, and if so an optimisation of them to allow the HEX to function more effectively. Another example is an analysis of the effect of changing the inlet velocity of the HEX to observe how the characteristics such as discharge time and thermal energy transfer change with this. Finally, an analysis of whether natural convection alone would provide a sufficient flow of air for the effective function of the HEX.

In domestic buildings, space and water heating is responsible for 80% of the total energy consumption [1]. To reduce this consumption, there is a necessity to improve the efficiency of space and water heating systems. Thermal energy storage (TES) is a possible solution to this by using a PCM HEX to store excess thermal energy for later use, this is possible due to their high heat storage capacity which is 5 to 14 times higher than sensible heat storage systems [2], making them a popular choice for TES [3, 4]. The use of these TES systems within both domestic and commercial building spaces allows for load shedding on the energy grid. This is because the TES device can be charged over a period in which the energy grid is under low demand or charged without the use of energy from the grid through methods such as solar power, geothermal energy or waste heat from other systems already installed within these buildings. This stored thermal energy can then be utilised for applications such as space or water heating at a time when the grid is under high demand. Utilising this stored energy reduces the demand on the energy grid which means it will experience less extreme fluctuation and therefore allows the grid to rely more on sustainable power solutions such as wind energy struggle to meet fluctuations in demand on their own.

This project will be using a PCM HEX that uses a wax PCM that is embedded within a metal foam. This form of PCM HEX is both effective and efficient for heating air due to the metal foam providing efficient thermal transfer routes. This massively enhances the performance of the PCM HEX and allows for a uniform temperature distribution as found in [5], foam with a porosity of 60% reducing the charging time by a factor of 100 and improving the thermal efficiency by up to 83%.

The current HEX sits at around 1.5m tall and weighs upwards of 100kg, a reduction in size and mass would be beneficial as it would improve handling and safety, but for this to be done the fans must be replaced with smaller alternatives or removed completely. If this is done it would bring another benefit to the HEX as reducing the fan size or removing it completely would reduce the chance of vibrations to the HEX therefore reducing the noise that it produces when in operation. This combination of mass, size and noise reduction would allow the HEX to become far more suited to application within dwelling spaces.

On top of that, the current design includes fins that stretch vertically up the HEX spaced at every 5mm. Currently, it's unknown whether these fins are beneficial to include, as they do improve the temperature profile and heat transfer to the air, but at the same time, they massively reduce the output velocity. If they are found to be effective, then another area worth

investigation is the optimisation of the spacing of the fins to transfer sufficient heat to the air without causing the PCM to discharge excessively fast.

The heating needs for dwelling spaces change constantly due to factors such as weather, time of day and season. For a heating system to be suitable for application in dwelling spaces it must be flexible to meet these needs which gives rise to a need to analyse the effect that velocity has on certain characteristics of the PCM HEX. The characteristics of interest are the discharge time and rate of thermal energy transfer as they will give a key insight into how well the HEX can be adapted to different heating needs. Following this, it would also be useful to analyse the performance of the PCM HEX using only natural convection compared to its performance with a fan. If natural convection alone allowed sufficient performance of the HEX it could allow for a reduction in production cost, mass, noise and running costs by removing the fan altogether, this would optimise the HEX for use within dwelling spaces.

This project is important as it will assess the current design of the PCM HEX and identify the areas of the system that require further analysis and optimisation to deliver an optimised solution that will implement the findings from the analysis. The project will also provide an insight into the characteristics of the PCM HEX under a range of inlet velocity scenarios including natural convection alone to determine its adaptability to a range of heating needs. There is currently a lot of literature on the internal workings of the PCM/foam but not much literature on the effect of the airflow system design such as the use of fins within the duct of the HEX [1]. There is also a lack of current literature on the effect the velocity of the air will have on characteristics of the HEX such as discharge time and thermal transfer rate. This project focuses on these areas currently underreported in the literature.

1.2. Aim and Objectives

Aim:

This project aims to enhance Heat Exchanger (HEX) efficiency by conducting a comprehensive assessment of fin impact through CFD analysis. This includes optimizing fin spacing for improved heat transfer and analysing the performance of Phase Change Material (PCM) HEX under varying inlet velocity conditions, including scenarios involving natural convection only. The overarching objective is to balance flexible airflow with optimized heat transfer rates and overall efficiency, while also gauging the suitability of PCM HEX for latent heat storage (LHS) applications within dwelling spaces.

Objectives:

- ☐ Evaluate the impact of incorporating fins into the HEX to enhance heat transfer efficiency. Utilize CFD simulations to compare HEX designs with and without fins, quantifying effects on airflow patterns, heat dissipation, and overall efficiency enhancements.
- ☐ Optimize the spacing of fins within the internal air duct to achieve optimal heat transfer to the air. Employ a combination of CFD simulations and numerical methods to analyse and validate results, ensuring optimal heat transfer performance.
- ☐ Investigate the influence of inlet velocity on HEX performance. Employ CFD simulations and mathematical calculations to assess thermal energy transfer and discharge times across a range of inlet velocities, providing insights into optimal operating conditions.
- ☐ Assess the suitability of the HEX for natural convection airflow scenarios. Utilize magnitude of order approximations and CFD models to analyse airflow characteristics and thermal performance, determining the feasibility of relying solely on natural convection for airflow through the duct.

2. LITERATURE REVIEW

PCM HEXs can be used to store thermal energy due to their high heat storage capacity which is 5 to 14 times higher than sensible heat storage systems [2], making them a popular choice for TES[3, 4]. The application of PCMs for TES has many benefits, they maintain a constant temperature during the phase change, PCMs are chemically stable, have no corrosion and can store large amounts of energy in a compact space [6], [7]. The need for such a system within dwelling spaces is growing due to the need for heating systems rising proportionately with population growth globally [8], and space and water heating is responsible for 80% of the total energy consumption in domestic buildings [1].

Within the literature, it is apparent that many investigations have taken place into the applications of PCM HEXs which include Absorption refrigeration systems [9], Waste heat recovery units in engines [10, 11], Photovoltaic cell systems [12-16], and central air conditioning systems [8, 17].

Based on this literature, it is clear that a lot of research has gone into the application of PCM for TES in large-scale systems and waste heat recovery within engines which shows that a common downfall of the PCM is the rate of heat transfer. This is a problem because it causes lengthy charging and discharging times of the PCM and reduces the HEX's ability to take full advantage of the thermal energy stored within the PCM [1, 5]. Most of the research involving PCM applications and PCMs themselves are based around the applications above which aim mostly to improve the thermal efficiency of systems or large-scale heating/cooling applications.

A large amount of research has been dedicated to combatting the problem with low rates of thermal energy transfer to/from the PCM. The Main focus of current and past research on the solution to this problem has been around Introducing porous metal foams to the PCM[18, 19], embedding nanoparticles within the PCM [6] [11, 20], adding fine conductive particles to the PCM [21, 22], the use of fins or baffles within the PCM [6, 23-25], the use of multiple PCM's [8] [12, 26], the physical geometry and arrangement of the airflow channels of the HEX [27], the geometry of the PCM [28-30], and the arrangement of heating elements [5].

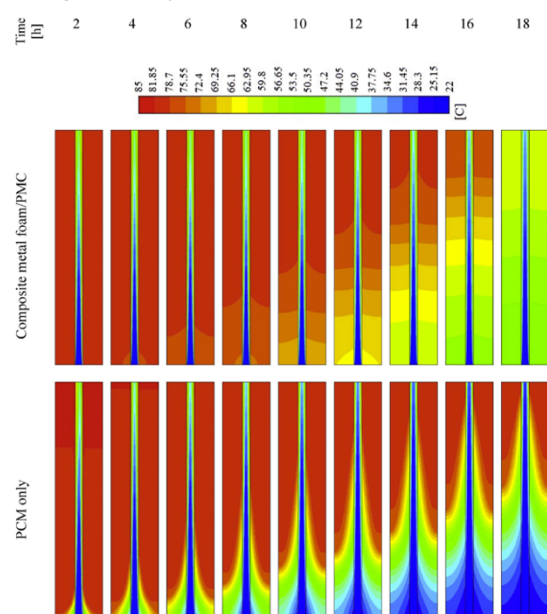


Figure 1: Comparison of Discharging with PCM only and Composite metal foam/PCM

All this research aims to increase our capability to efficiently access the large amounts of latent heat stored within the PCM at a desirable rate based on the application at hand. One key paper presents the method of embedding metal foam within the PCM. This method has many advantages over other techniques as it promotes a uniform melting process across the PCM whilst providing greatly increased heat transfer, this is demonstrated clearly in [1] through analysis shown in Figure 1. The simulations conducted in this paper are done on a PCM HEX that is similar to the one in this project so it will be key in the verification of simulations completed and providing the thermal behaviour of the PCM/metal foam to use for this project's simulations.

Another paper of key relevance to this project is [27], which ran simulations to test various types of duct geometry changing both width and airflow path to increase thermal output. It was found that a straight duct is the most effective over the whole melting period and that the width of the duct does not have as large of an effect on thermal output as geometry. This will play a significant role in producing the simulations for the effectiveness of the fins work package as some of the simulations ran within [27] are using a similar HEX model to the one in this project.

The information found in the literature on the applications and improvement of thermal energy transfer for the PCM will be of great use to this project as the design of the PCM HEX is based around a PCM embedded within a porous metal foam. The use of such knowledge gives a strong foundation to build upon and will allow the project to be focused on areas not yet covered by literature. On top of that, it will be possible to use the research found on the airflow channel arrangement to help verify the simulations of this project as they present the behaviours that a similar PCM HEX setup exhibits.

The findings of this review show that great investigation has gone into applications and thermal efficiency improvements, but some areas of this investigation lack insight. One of these areas is the use of PCM HEX within dwelling spaces, this has been overlooked as a lot of the research conducted was on much more large-scale systems. The application of PCM HEX for TES could be an effective solution to improving energy efficiency in dwelling spaces so there is a clear void in the current research that could provide valuable insight into an overlooked application.

Furthermore, there appears to be a lack of research into the use of fins within the ducts of a PCM HEX. The use of them within the PCM itself has been investigated [6, 23-25], but the effect of introducing fins into the duct where the heat transfer takes place has not been investigated. An additional area that had a clear lack in literature was how the inlet velocity of the air affects characteristics of the PCM HEX such as discharge time and thermal energy transfer rate. This research could provide valuable information when assessing the suitability of the PCM HEX for use within dwelling spaces.

These areas missing research open some important points that this project hopes to cover, such as how the use of fins within the duct affects the heat transfer to the air and the effectiveness of the heat exchanger and what would be the optimal spacing of the fins to make best use of the LES provided by the PCM. Also, how could variations in velocity be utilised to enhance the PCM HEXs performance within dwelling space applications to provide high-efficiency solutions to low-energy heating.

3. METHODOLOGY

This Project consists of three work packages that build on one another to optimise and gain a better understanding of the PCM HEX. These work packages are Analysis of the effectiveness of fins, optimisation of fin spacing, and analysis of the effect of inlet velocity. Each of these work packages has been completed with a different methodology to allow their experiments and findings to be tailored to their individual needs. When one of the work packages is complete the findings are used to begin the next work package for example the results from the optimal fin spacing will be used to complete the analysis of the effect of velocity, doing this provides both the optimal fin spacing and a deeper understanding of the effects this has and the performance of the HEX with the optimised spacing in place.

3.1. Methodology of Experiments

Analysis of the Effectiveness of Fins

To begin a CFD analysis on the effectiveness of incorporating fins into the air duct of the HEX was completed. To do this only the duct was modelled with the use of suitable boundary conditions to represent the insulating and heating components of the HEX allowing the problem to be simplified so detailed results could easily be obtained for the HEX without fins. To model the HEX with fins a similar process was followed as the fins turn the duct within the HEX into a series of much smaller ducts, only one of these ducts was modelled as it is expected that all the ducts exhibit the same behaviour. These simplifications were necessary to obtain accurate results as the ANSYS student license has a limit of one million nodes, so a smaller model is required to use a dense mesh. After the simulations had been completed figures were produced to visually represent the data through planes as well as rakes being taken and plotted with the use of MATLAB scripts to represent temperature profiles. These figures were then used to perform a detailed analysis of the results allowing conclusions to be made.

Optimisation of Fins Analysis

The optimisation of the fin spacing work package aims to obtain the optimal spacing of the fins considering the heat transferred to the air and pressure drop due to airflow resistance. This optimisation also has the potential to reduce the mass and manufacturing costs of the PCM HEX and potentially allow the use of a cheaper and more compact fan. To do this a more detailed set of CFD simulations was used which contained a whole slice of the HEX duct including the material of the fins in the simulation and the effect of congregate heat transfer. Mathematical calculations were used to validate the CFD simulations setup and methodology by using the one-dimensional heat conduction equation to calculate the theoretical temperatures along the fin and comparing these values to values obtained within the CFD simulations.

To find the optimal distance for the fin spacing two sets of simulations were conducted. The first was a coarse set of simulations conducted that were run on fin spacings of 5mm to 25mm in 5mm increments. For each of these increments' rakes were used to export temperature profiles throughout the duct and then plotted using MATLAB to give clear and precise results for analysis, doing this will give insight into what range of spacings holds the optimum value

and give the range of spacings that the second more refined set of simulations will cover. This more refined set of simulations was conducted with 1mm increments to find the optimal spacing of the fins with the same detailed CFD methodology that will be used as the coarse set of simulations. As before for each case rakes will be used to export and plot temperature profiles along the duct but this time pressure profiles will also be exported and plotted using MATLAB to give insight into the pressure drop in the duct showing the changes in resistance to airflow. From the analysis of the results and generated figures, conclusions were made on the optimal spacing of the fins and the effects this will have on the thermodynamic behaviour of the PCM HEX and the behaviour of the airflow through the duct.

Effect of Velocity Analysis

The effect of the velocity work package is aimed at analysing the effect that the inlet velocity has on different characteristics of the HEX such as the rate of thermal transfer, the discharge time of the PCM, and the temperature profile. This will give a key insight into both the performance of the PCM HEX under the optimisations that have been made and a better understanding of the ability to implement such systems into application for heating and energy saving within dwelling spaces. To do this the CFD model for the optimal case in the optimisation of the fins work package was used as this had already been verified to be accurate and includes a slice of the HEX with the optimal fin spacing implemented. There was then a set of simulations ran on this model with a range of input velocities going from $0.25 \frac{m}{s}$ to $1.5 \frac{m}{s}$ which will provide a suitable range of data allowing conclusions to be gathered.

Once these simulations have been completed the data is exported using rakes to be imported into Excel allowing calculations to be performed. Firstly, the thermal energy transfer was calculated using the inlet and outlet temperatures obtained from ANSYS, the thermal energy transfer was then used to calculate the discharge times of the HEX at the different inlet velocities as it is known that the PCM within this HEX holds 5kWh of thermal energy. The thermal energy transfer and discharge times were then both plotted onto their own graphs to allow for analysis of the trends within them. On top of that the rakes were used to plot the temperature profiles of the air across the duct as this will provide a useful insight into how changing the velocity of the air compares to changing the spacing of the fins.

Finally, an analysis will be performed to understand how the HEX performs by using natural convection only. This will be done by first calculating an approximate order of magnitude for the velocity caused by natural convection. Following this, a simulation will be set up using the same model as all the other simulations in this experiment but this time setting the inlet velocity to the one found in the previous calculation. From this, the data will be exported allowing analysis of how the HEX performs using natural convection only giving information on if the fan is truly necessary or the thermal transfer can be controlled by opening / restricting the airflow.

3.2. CFD Model Setup

Analysis of the Effectiveness of Fins

To complete the effectiveness of the fins analysis work package two simulations have been run through Ansys using the Fluent solver. The first simulation modelled the airflow between the internal fins within the HEX to obtain temperature and velocity profiles to assess the performance of including fins. To set up the solver 4 CPU cores were used, double precision was enabled, k-epsilon and standard wall treatment were used, the energy equation was turned on, gravity was applied at $-9.81 \frac{m}{s^2}$ to the y-axis and the air density was set to behave as an ideal gas to account for buoyancy and compressibility. A standard wall function was used along with the k-epsilon viscous model and laminar flow is expected. To set up the mesh for the model a hex dominant method was used due to the simple rectangular geometry which can be seen in Figure 2. An element size of 0.5mm was used to ensure accuracy while keeping within the Ansys student node limit of one million. Finally, boundary conditions were added to the simulation, a temperature boundary condition of $75^{\circ}C$ was added to the side walls of the duct as this accurately represents the thermal output from the HEX. Inlet and outlet boundary conditions were also used for the inlet the boundary condition was a velocity inlet which was set to 0.5 m/s due to this being a generally used value used in other simulations with a similar aim [31], and the outlet boundary condition was a pressure outlet set to 0bar (gauge).

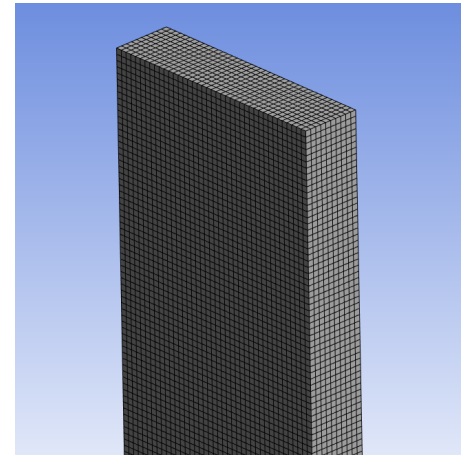


Figure 2: Mesh of Air Channel Between Fins

For the second simulation in the effectiveness of fins analysis work package the same solver settings, viscous model and wall function were used as the first simulation in this work package and laminar flow is expected. The mesh and boundary conditions were changed along with the setup of the model itself to improve accuracy. This simulation modelled the whole duct as the aim was to assess the performance of the HEX when no fins were included. To set up the mesh for the model a hex dominant method was used due to the simple rectangular geometry which can be seen in Figure 3. An element size of 0.01m was used to ensure accuracy while keeping within the Ansys student node limit of one million. The large side walls boundary conditions were set to $75^{\circ}C$ representing the thermal energy provided by the PCM and the side walls were insulated to represent the existing HEX prototype. The inlet and outlet boundary conditions for this simulation were the same as the first simulation in this work package to keep continuity between simulations.

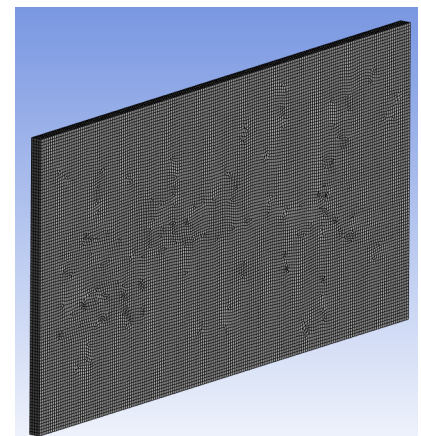


Figure 3: Mesh of Air Channel Without Fins

Optimisation of Fins Analysis

To complete the optimisation of the fin spacing work package two sets of CFD simulations were used but all simulations used the same solver and settings with the only difference being the number of nodes due to the changes in geometry with each simulation. The geometry of this CFD model is a slice of the HEX duct that contains a whole duct with 2 fins and half of a duct on either side, this is to ensure the behaviour of both the fins and the duct are being modelled accurately. These CFD simulations have been run through Ansys using the Fluent solver with the SIMPLE algorithm. To set up the solver 4 CPU cores were used, double precision was enabled, Reynolds stress and standard wall treatment were used, the energy equation was turned on, gravity was applied at $-9.81 \frac{m}{s^2}$ to the y-axis, the air density was set to behave as an ideal gas to account for buoyancy and compressibility. A standard wall function was used along with the Reynolds stress viscous model and laminar flow is expected. The meshing was done by using a global element size of 0.7mm and sweeps were used on the fins and the duct to ensure that there were at least 7 elements across the width of the fins and the duct to model the heat transfer accurately this can be seen in Figure 4. The sides of the fins and the duct (on the left of Figure 4) were set to a temperature of $75^{\circ}C$ this simulates the heat produced by the PCM and allow heat transfer to the fins and the air. The mesh interface between the fins and the air was set as a coupled wall. A symmetry boundary condition is also used on the sides of the half duct (on the right of Figure 4) as this allows a larger portion of the HEX to be simulated whilst making more efficient use of the node limit of one million. Inlet and outlet boundary conditions were also used for the inlet the boundary condition was a velocity inlet which was set to 0.5 m/s due to this being a generally used value used in other simulations with a similar aim[31], and the outlet boundary condition was a pressure outlet set to 0bar (gauge).

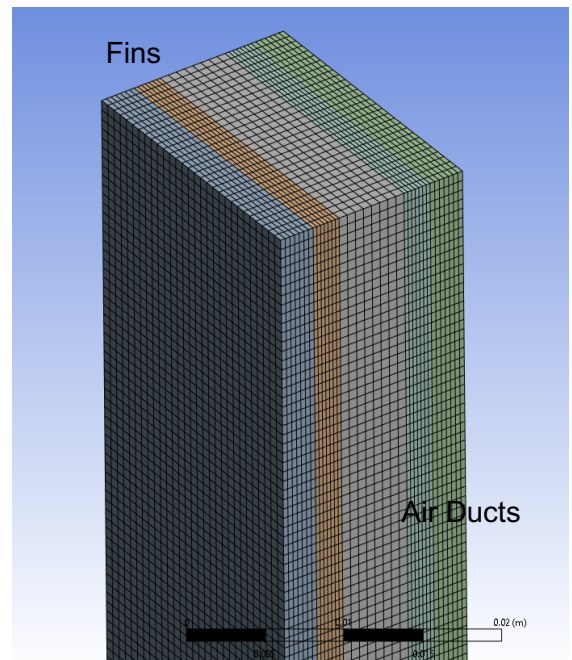


Figure 4: Mesh of HEX slice with fin and duct.

Effect of Velocity Analysis

To complete the effect of velocity analysis a single CFD model was used for a set of simulations in which the inlet velocity boundary condition was increased in increments of $0.25 \frac{m}{s}$ over the range of $0.25 \frac{m}{s}$ to $1.5 \frac{m}{s}$ with all other settings, geometries and meshes remaining the same throughout the set. For this, the same setup, geometry, and mesh were used as the optimal case found in the optimisation of the fins work package. Which included the use of the SIMPLE algorithm, the energy equation, gravity to be applied to the Y-axis and the air density set to behave as an ideal gas. The solver used was again the Reynolds stress viscous model with standard wall function and laminar flow is expected. Again, the boundary conditions were mostly the same with symmetry being used to allow for a higher density mesh and the HEX walls being set to a temperature of $75^{\circ}C$ with the only change being the velocity of the inlet boundary condition which changes for each simulation in this set as previously

mentioned. The reason behind following the previous simulation as closely as possible is because it was previously verified with mathematical calculations giving confidence in the accuracy and validity of the results obtained from this final work package on the effect of velocity, also it saves time to re-use a tried and tested method than to make a new one allowing more research to be conducted in the limited timeframe.

As previously stated, the natural convection case uses the same CFD setup as the other simulations in the effect of velocity analysis. The only difference is that the inlet velocity will be set to the value that was found in the upcoming calculation.

4. Validation

4.1. Optimisation of Fins Analysis

To ensure that the results obtained through the simulations in the optimisation of fin spacing work package are accurate and reliable mathematical calculations were used to validate the results of these simulations. For this, the one-dimensional heat conduction equation was used assuming steady-state conditions and neglecting heat generation within the fin and. Within this half of the fin was calculated and boundary conditions were used to represent an insulated tip which means that there is no heat transfer at the tip, so the temperature gradient $\left(\frac{dT}{dx}\right)$ at the tip is zero. To do MATLAB was used to plot Equation 1 with the boundary conditions and suitable input values.

Once this had been completed Figure 5 was produced to present the values calculated within MATLAB. This graph presents the temperature of the centre of the fin from the PCM HEX (0m) wall to the centre of the duct (0.01m), it is important to note that the results are symmetrical about the centre of the duct hence why only one half has been calculated. Looking at the graph the fin starts at the HEX wall with a temperature of 75°C and the temperature drops at a constant rate until it reaches the centre of the fin where it is at a temperature of 62.3°C gives an overall temperature drop of 12.7°C . This temperature drop is caused by convective heat transfer to air passing over the fin.

To compare these theoretical values to the ones calculated within ANSYS fluent a rake was used to export temperature data at 40 points along the width of the fin through the centre line. MATLAB was then used to plot this exported data which is presented in Figure 6 note that this time the temperature profile is along the whole fin (HEX wall to HEX wall) and it can be seen that the temperature profile is symmetric as expected. Looking at the graph the fin starts at the HEX wall with a temperature of 74.5°C and the temperature drops at a constant rate until it reaches the centre of the fin where it is at a temperature of 68.8°C giving an overall temperature drop of 5.2°C .

When comparing the values obtained through mathematical calculations and the ones produced through CFD simulation there is a difference between them. The mathematical values have a temperature drop along the fin of 12.7°C and the CFD temperatures have a drop

of 5.2°C along the fin. The main reason behind this difference is the fact that the mathematical calculation is a one-dimensional approximation that will not consider the convective heat transfer from the vertical component of the fin. The CFD model does take this into account as it models the entire fin and duct in a 3D space, this would give the 3D model smaller temperature drops along the fin accounting for the difference that we can see. Another reason for the discrepancy is that the mathematical calculation only uses one equation and the CFD simulation uses seven due to the use of the Reynolds stress model which would give a more accurate solution to the system.

Overall, the mathematical calculations validate the method of the CFD simulation as the discrepancies can be expected due to the properties of the two types of calculations and the temperature profiles are within a suitable tolerance of each other due to the factors discussed. This means that the optimisation process is in a suitable position to move forward to produce accurate results on the optimal spacing of the fins for the PCM HEX.

Equation 1: One-Dimensional Heat Conduction Equation

$$\frac{d^2T}{dx^2} = -\frac{h \times P}{k \times A} \times (T - T_{\infty})$$

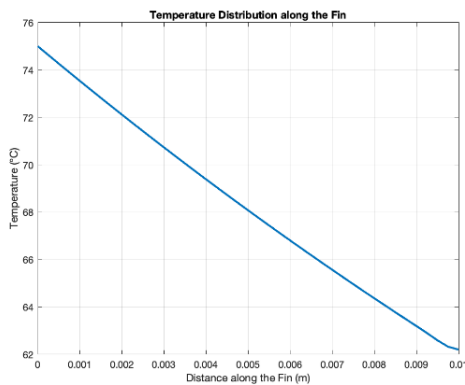


Figure 5: Temperature Distribution Along the Fin (MATLAB)

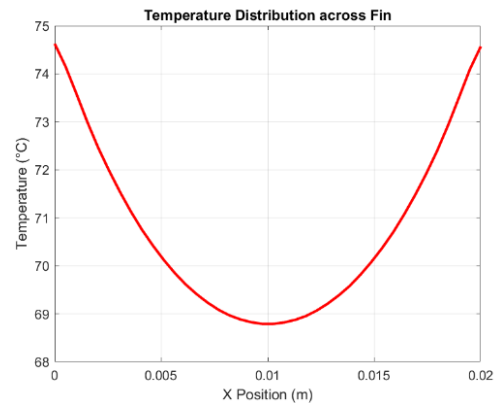


Figure 6: Temperature Distribution Along the Fin (Ansys Fluent)

5. RESULTS AND DISCUSSION

5.1. Effectiveness of Fins Analysis

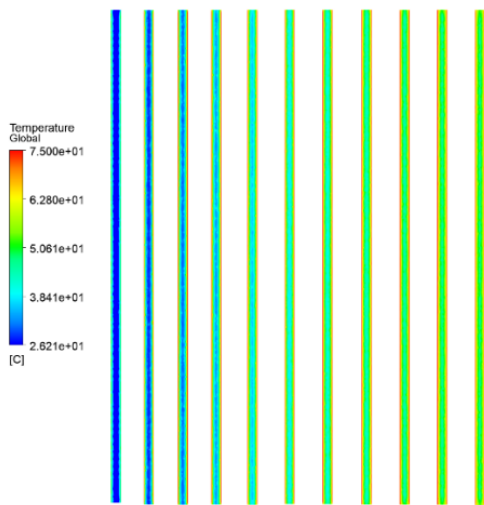


Figure 7 : HEX Without Fins Horizontal Temp Profiles

After both simulations for the effectiveness of the fins analysis work package had been completed, an initial visual comparison was completed. This was done the same way for both simulations by generating a series of planes across that stretch across the duct showing the temperature profile at vertical intervals of 0.05m. Looking at the HEX without fins in Figure 7, we can see that the average temperature of the duct goes from about 26 degrees C to about 55-60 degrees from the entrance (far left) to the exit (far right). But at the exit, there is a large variation between the maximum temperature and the minimum temperature, 70 and 55 degrees C respectively.

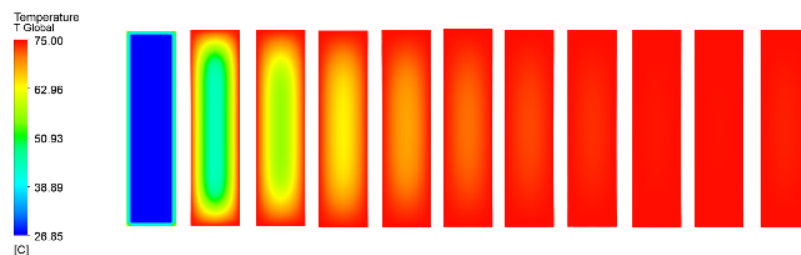


Figure 8 : HEX With Fins Horizontal Temperature Profiles

Moving to the results of the HEX with fins that are presented in Figure 8 we can see that the temperature profile changes quickly from the inlet temperature of 26 degrees C (far left) to the outlet temperature of 75 degrees C (far right). The air is quickly heated to the outlet temperature where it reaches this and remains constant after 0.3m through the duct after which no further heating is taking place.

After looking at the results presented in Figure 7 and Figure 8 we have a basic understanding of what is going on but a more detailed comparison will need to be made to draw any conclusions.

To do this MATLAB was used to plot data that had been collected by using rakes in Ansys, these rakes were taken at the same height intervals as the planes previously presented and

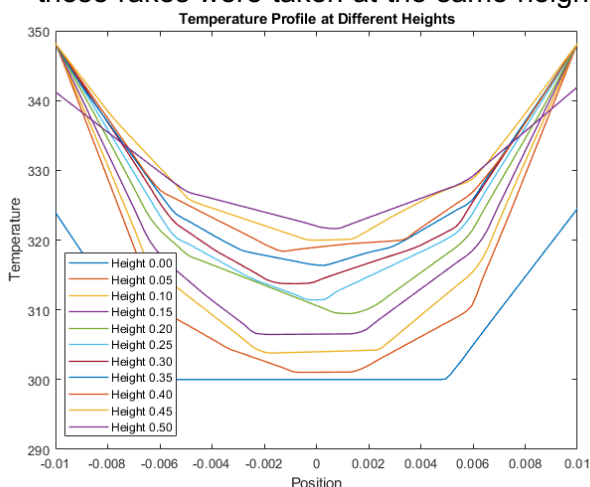


Figure 9 : Temprature Profile of HEX Without Fins

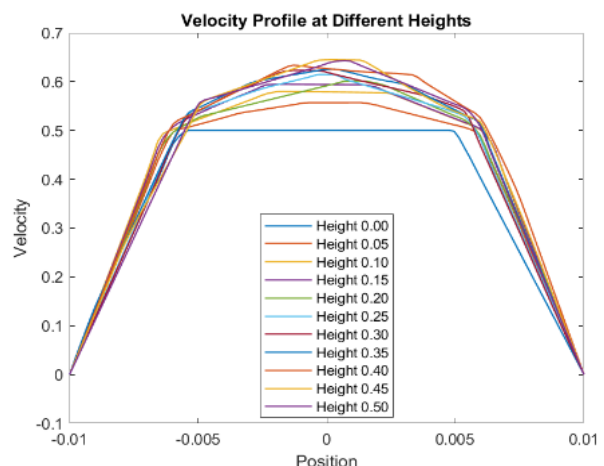


Figure 10 : Velocity Profile of HEX Without Fins

they stretched across the centre of the duct going from the HEX wall to HEX wall, extracting both temperature and velocity data for each simulation. In Figure 9 the temperature profile of the HEX without fins is presented and verifies the previous points made from the visual assessment of the temperature planes. We can see that the temperature profile is always higher when near to the HEX walls causing the output temperature to have a large variation of 50 degrees C at the centre and 75 degrees C at the HEX walls. This is an undesirable characteristic as we are looking for a uniform output temperature. It can also be seen that the velocity profile for the HEX without fins presented in Figure 10 we can see that the air comes in at 0.5 m/s and then accelerates up to about 0.65 m/s, this acceleration will be caused by natural convection due to the air gaining thermal energy.

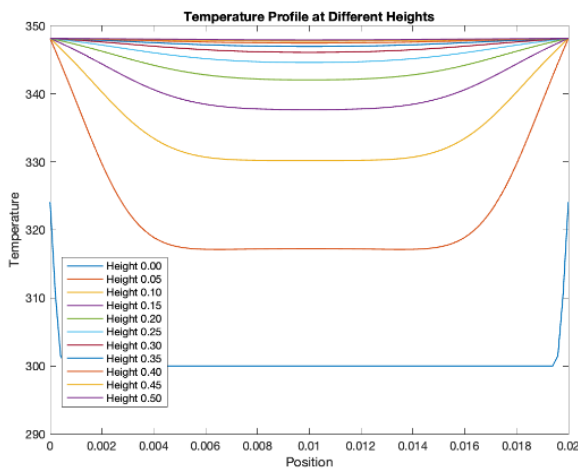


Figure 11 : Temperature Profile of HEX With Fins

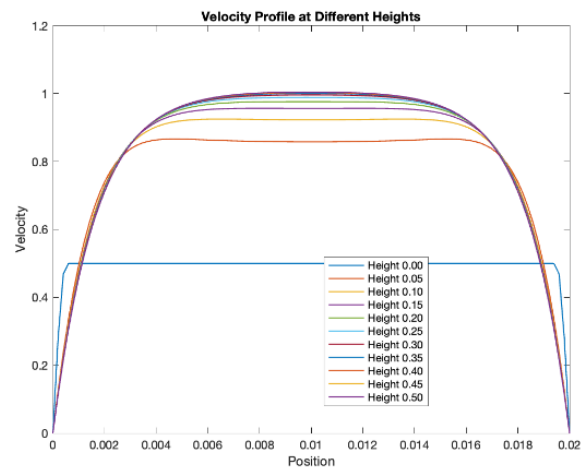


Figure 12 : Velocity Profile of HEX With Fins

Presented in Figure 11 is the temperature profile of the HEX with fins running through the duct. This again confirms the statements previously made from the temperature planes. We can see that the temperature profile at the outlet of the HEX is uniform but from 0.3m onwards it only gains a small amount of thermal energy. This indicates that the fin spacing is too small causing thermal energy to be transferred to the air at an unnecessary speed which will drain the PCM too quickly and cause uneven solidification. A way to improve this could be to increase the spacing of the fins to find an optimal value that will allow the air to reach a uniform output temperature closer to the outlet of the HEX. The velocity profile shown in Figure 12 shows that the fins did not have a negative effect on the airflow, but they increased it this would be down to the effect of natural convection.

Overall, the results presented show that not using fins causes the HEX to have a non-uniform output temperature whilst not getting the maximum amount of thermal energy available. When the fins were introduced to the HEX it was demonstrated that they significantly enhanced the heat transfer to the air. With the fins, the heat transfer was occurring too quickly and only near the inlet of the HEX not making use of the full length of the duct which would cause the PCM to melt unevenly. From this, an investigation is required into the optimal spacing of the fins within the PCM HEX to make the improvements stated and these optimised fins will then be able to be used in the final design.

5.2. Optimisation of Fins Analysis

Optimisation of Fin Spacing - Rough test

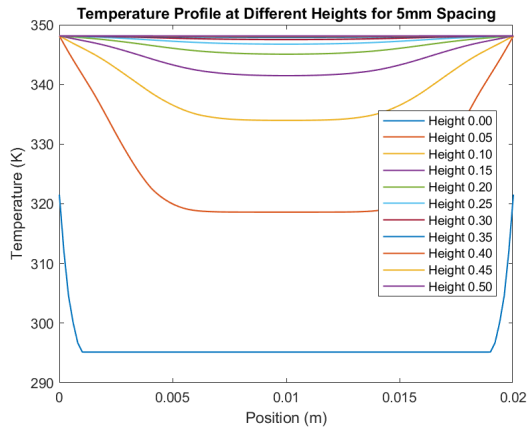


Figure 13: Air Temperature Profiles Throughout Duct For 5mm Spacing.

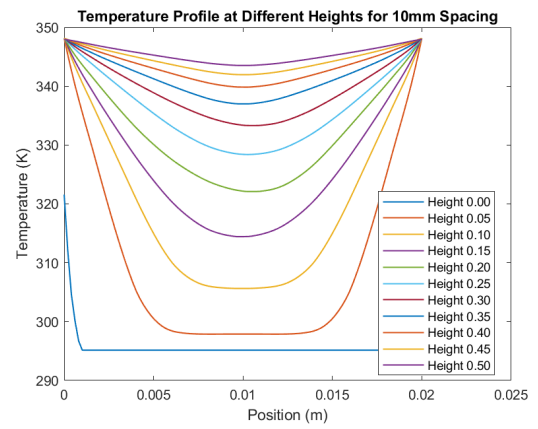


Figure 14: Air Temperature Profiles Throughout Duct For 10mm Spacing.

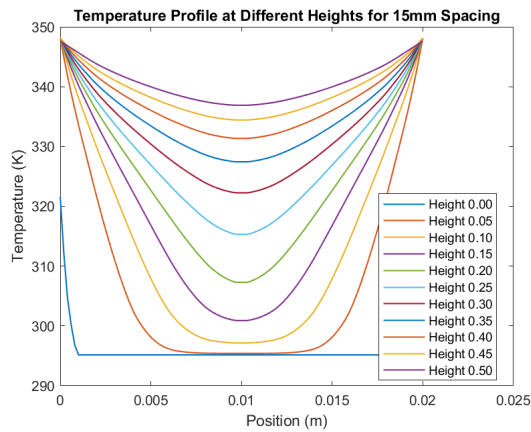


Figure 15: Air Temperature Profiles Throughout Duct For 15mm Spacing.

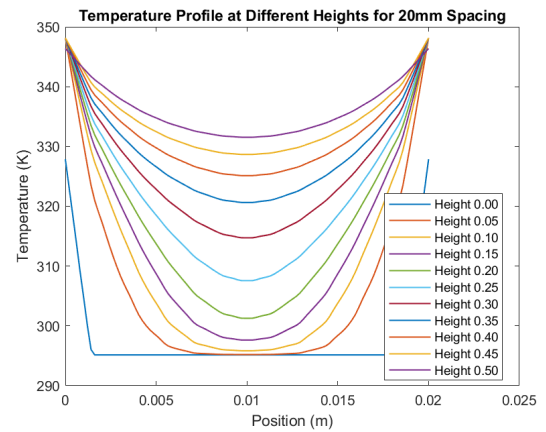


Figure 16: Air Temperature Profiles Throughout Duct For 20mm Spacing.

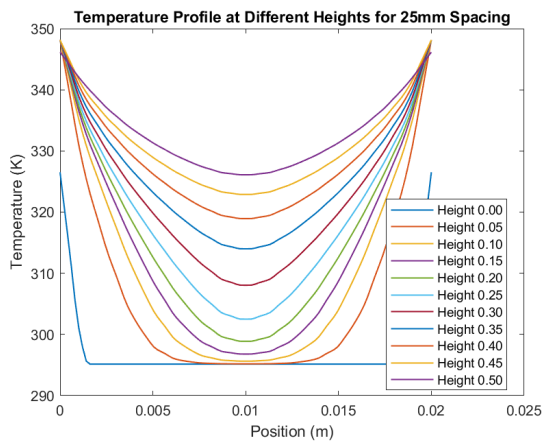


Figure 17: Air Temperature Profiles Throughout Duct For 25mm Spacing.

As discussed in the method the optimisation begins with a set of simulations that have 5mm increments of fin spacing between 5mm and 25mm. The results for these CFD simulations are presented in Figure 13 to Figure 17, each of these figures contains the temperature profiles taken at the centre of the fins going across the duct from HEX wall to HEX wall with a profile being taken in 50mm increments up the 500mm duct. This allows for the analysis of the development and changes in the temperature profile through the duct to get a clear understanding of the thermal transfer taking place.

When comparing the results between the conducted simulations the temperature profiles are the main factor in deciding what range would be most suitable for a more refined investigation. Two main factors contribute to a good temperature profile for this application. Firstly, the air must be gaining thermal energy all the way up the duct as this will allow for a more even solidification of the PCM which helps make the most efficient and effective use of the charge stored within it and will give a more predictable temperature outlet when the PCM HEX is in use. Also, an even temperature outlet that is close to the temperature of the HEX walls is important as this shows that the air has gained as much thermal energy as possible to be delivered to the surrounding air and raise the room temperature at a fast rate.

Comparison of 5mm and 25mm cases

Looking at the presented figures for this experiment it can be seen that the best comparison to be made showing the effect of the change in fin spacing is between Figure 13 (5mm fin spacing case) and Figure 17 (25mm fin spacing case), both of these cases are not ideal for the opposite reasons. Firstly, the 5mm case shown in Figure 13 is not ideal due to the air in the duct reaching the maximum temperature 0.3m up the 0.5m duct. This means that for the last 0.2m, no thermal transfer is taking place so the first 0.3m of the PCM will solidify much quicker than the rest of the PCM. In the 25mm case shown in Figure 17 it can be seen that the air does not reach the wall temperature of 348 K as the air in the centre of the duct is only at 327 K so not enough thermal transfer is taking place. Overall, in the 5mm case, the thermal transfer is happening too quickly to make full use of the HEX and in the 25mm case, not enough thermal transfer is taking place to deliver sufficient energy to the surrounding room. From looking at these comparisons it can be noted that the optimal fin spacing will be between 5mm and 25mm as the 5mm spacing is too small and the 25mm spacing is too large.

Eliminating 15mm and 20mm cases

Looking at the figures with the points of comparison previously discussed allows two more increments of spacings to be noted as sub-optimal which are the 15mm and 20mm cases shown in Figure 15 and Figure 16 respectively. The fin spacing in both cases is too large and has the same issues as the 25mm with not enough thermal energy being transferred to the air. Although, it is important to note that as the fin spacing is getting progressively smaller in these cases more thermal energy is being transferred to the air and the temperature profiles are slowly developing to meet the requirements.

Discussion of 10mm case

Finally, looking at the 10mm case shown in Figure 14 this case is the most optimal of the set due to the air gaining thermal energy along the entire duct whilst getting close to the maximum outlet temperature along a relatively flat temperature profile. This means that the air travelling through the HEX is making effective use of the length of the duct which will help the PCM solidify evenly and allow the air to gain sufficient thermal energy to effectively heat a room. Although this is the most optimal out of this set of results there is room for further optimisation

as there is still variation along the temperature profile, this further optimisation would involve using a slightly smaller fin spacing to achieve a higher outlet temperature across the duct.

Selecting the refined test spacing range

Using the results presented and discussed in this stage of the optimisation it has become apparent that the next stage will be conducted between the 5mm spacing and the 10mm spacing. The reason for this is due to the 10mm spacing being the most optimal of the current stage but there is still room for further optimisation on getting a higher average output temperature which would be obtained by reducing the fin spacing. On top of that there needs to be an analysis of the effect that the changes in fin spacing will have on the pressure drop experienced across the duct doing this will allow a more informed decision to be made on the optimal fin spacing when a range of spacings that have suitable thermal behaviours have been found in the upcoming refined optimisation.

Optimisation of Fin Spacing - Refined Test

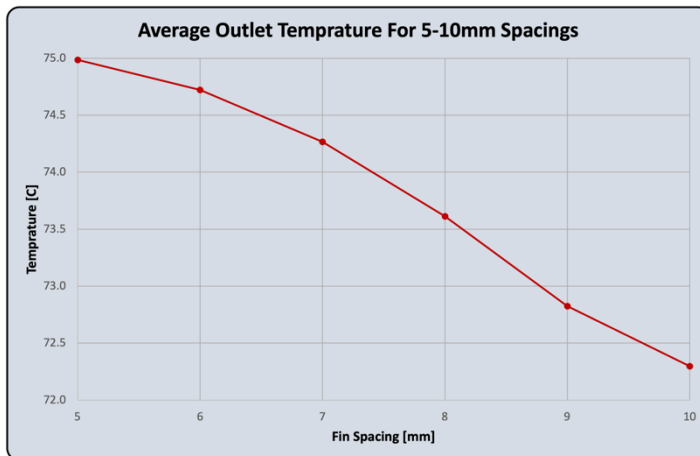


Figure 18: Average Outlet Temperature for 5-10mm Fin Spacing

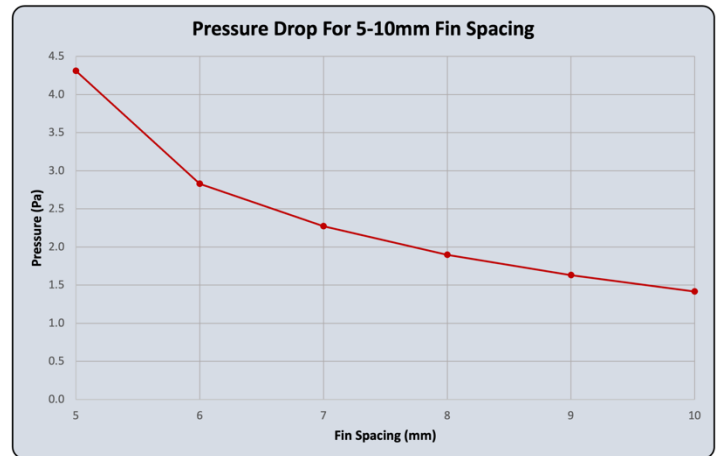


Figure 19: Pressure Drop Along Duct for 5-10mm Fin Spacing

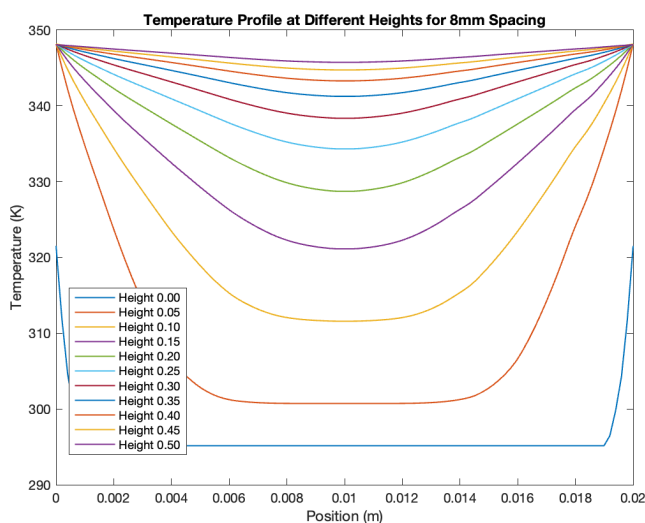


Figure 20: Air Temperature Profiles Throughout Duct For 8mm Spacing.

The refined stage of this optimisation process is run with a similar method to the previous stage but this time the simulations are conducted with fin spacings of 5mm to 10mm in 1mm increments which will allow a refined optimal fin spacing to be found. To allow for effective analysis of each spacing the average outlet temperature, pressure drop along the duct and temperature profile will be considered these are presented in figures Figure 18 to Figure 20 respectively. Temperature profiles have been taken with the same method as the previous stage for each increment of fin spacing but only the

8mm one has been presented here, the others have been presented in the appendix.

Temperature outlet analysis

The average outlet temperature of the air for each of the simulations is presented in Figure 18, as expected there is a trend that as the fin spacing grows the temperature of the air coming out of the duct will drop. Looking at Figure 18 a 5mm fin spacing the average air temperature is at the HEX wall temperature of 75°C compared to a 10mm fin spacing where the average outlet temperature is 72.3°C it can be noted that there is not a significant drop between these two values but the higher the temperature would always be the most ideal.

Temperature profile analysis

To get a better insight into this the temperature profiles were compared as before (these can be found in the appendix), and it was noted that the 5mm and 6mm spacings both had the same problem. This was that the air reaching the maximum temperature before the end of the duct which would cause uneven solidification of the PCM which as stated previously is a sub-optimal situation. With this information, it can be stated that the 7mm or 8mm cases would be preferred when comparing the thermal results of the fin spacings as they use the whole length of the HEX to gain thermal energy whilst still reaching a suitable temperature this can be seen in Figure 18 and Figure 20.

Pressure drop analysis

The average pressure drop across the duct for each of the simulations is presented in Figure 19. The pressure drop displays a similar trend to the average temperature as the fin spacing widens the pressure drop decreases. The reason that the pressure drop is decreasing is due to the reduction in friction as the air has more space to pass through the duct at the same speed. The pressure drop reduces the most when going from 5mm to 6mm fin spacing and then the reduction in pressure drop gets smaller in each of the following increments. Ideally, the optimal fin spacing will need to have the lowest possible pressure drop.

A reduction in pressure drop will bring multiple benefits to the design of the PCM HEX. Firstly, it will mean a smaller less powerful and expensive fan will be suitable for this application as the air requires less pressure to be moved at the same rate through the duct. This smaller fan would also reduce the overall weight of the PCM HEX and the overall size improving the packaging of the design. The increased spacing of the fins will also mean a reduction in manufacturing costs and mass as fewer fins will be used to produce the duct. The combination of these will help make the PCM HEX more suitable for use within dwelling spaces as their main issues are their size, mass and the cost of manufacturing making them a less practical option for space heating.

Selection of optimal spacing

After analysing both the thermal behaviour and pressure drops for each of the iterations of this final stage of the optimisation process it was found that the 8mm fin spacing is the most ideal solution. One of the reasons for this is because the 8mm spacing used the whole length of the duct to gain thermal energy and then the air exiting the duct had an even temperature profile

that was at a suitably high temperature. This shows that the 8mm fin spacing has ideal thermal behaviour for application within this PCM HEX. Another reason the 8mm spacing was chosen over the 7mm fin spacing as they were both suitable when comparing the thermodynamic behaviour was due to the 8mm spacing having a greater pressure drop, the pressure drop for the 8mm case was under half of the value from the 5mm fin spacing that the previous design had.

Overall, this optimisation process has been a success as it has met its aim of finding the optimal fin spacing of 8mm for this PCM HEX. This optimisation will provide improvements to the design and efficiency of the HEX. The PCM will solidify more evenly along the duct as the whole length is now used for heat transfer, the reduction in fins will lower the overall mass and reduce manufacturing costs, and the reduction in pressure drop will allow for a smaller less powerful fan that will allow for a reduction in noise, mass, and production cost.

5.3. Effect of Velocity Analysis

Calculations

To perform a thorough analysis of the effect of velocity it was necessary to perform calculations with the use of data exported from the simulations completed in ANSYS. These values were the inlet velocity of the air, the inlet temperature, and the outlet temperature. Along with these values, the specific heat capacity, Inlet area and density were required along with the thermal capacity of the PCM within the HEX. The values for each of these are contained within Table 1 using their corresponding symbols which can be found in the nomenclature section of this report.

VARIABLE	VALUE	UNITS			
C_p	1.005	$\frac{kJ}{kgK}$	Mass flow =	$\dot{m} = \rho AV$	Equation 2
A_i	0.016	m^2	Thermal energy Equation =	$\dot{Q} = \dot{m} c_p (\Delta T)$	Equation 3
ρ	1.2	$\frac{kg}{m^3}$	Discharge time =	$t = \frac{Q}{\dot{Q}}$	Equation 4
Q	5	kWh			

Table 1: Values used in calculations.

Firstly, the mass flow rate of the entire HEX for each inlet velocity was calculated by using Equation 2 which then allowed the amount of thermal energy is transferred to the air for each inlet velocity to be calculated through equation with the use of C_p and ΔT . To obtain ΔT it was necessary to export rakes of temperature values across the duct outlet and then calculate the average values to get T_{out} , and T_{in} was already known as a boundary condition. Bringing all this together with Equation 3 to obtain the thermal energy transfer to the air, this is then plotted on a graph within Excel to allow for analysis into the effect that velocity has on the energy transferred to the air.

Another important calculation to be made was the discharge time as this will allow an assessment to be made on whether simply changing the velocity of the air would be enough to regulate the discharge time of the HEX to operate over a designated period like the way modern heating systems only operate over designated times. To do this Equation 4 was used which gives the time in hours by simply dividing the thermal capacity of the HEX by the amount of energy that is taken by the air. The results of this are again plotted onto a graph to make it visible the effects that the change in velocity has on the discharge time and allow trends to be found.

To analyse how the HEX would perform using natural convection only it was necessary to perform calculations to estimate the velocity of the air flowing through the HEX. This is due to the limitations of ANSYS not being able to reliably run an open system natural convection case. To conduct this calculation two formulas were needed which are presented below, Equation 5 is used to calculate the density of the air this is done at the start of the HEX and the end of the HEX based on inlet thermal boundary conditions and estimated outlet temperatures. After doing this it is possible to then use Equation 6 which was obtained from [32] to find the approximate order of magnitude of the velocity. To ensure accuracy this velocity was then used to interpolate an outlet temperature from the set of simulations previously conducted over a range of velocities, and then the calculation was performed again with this updated value to obtain a more accurate solution. The input variables used are presented in Table 2 along with their units.

VARIABLE	VALUE	UNITS	
P	101	kPa	Density =
R	8.314	$J/mol\ K$	$\rho = \frac{P}{RT}$ Equation 5
T_{in}	22	$^{\circ}C$	Velocity order of magnitude =
g	9.81	$\frac{m}{s^2}$	$V \approx \sqrt{\frac{\Delta\rho}{\rho} gL}$ Equation 6
L	0.5	m	

Table 2: Values used in Velocity

After completing several iterations of this calculation to ensure accurate results the velocity was found to be $0.843 \frac{m}{s}$. As stated, this is only an order of magnitude approximation so the result will not be entirely accurate due to the approximation being of a flat wall and not a duct. This will still provide a suitable insight into how the HEX will perform using natural convection only and determine whether it is possible to function properly without the use of a fan.

Variations of Inlet Velocity

As discussed in the method section the analysis of the effect of velocity consists of a set of simulations each with different velocities ranging from $0.25 \frac{m}{s}$ to $1.5 \frac{m}{s}$ with increments of $0.25 \frac{m}{s}$. The Results for these simulations have been exported and compiled into figures for analysis which are presented in Figure 21 to Figure 23. The goal of this is to get a clear understanding of the characteristics of the optimised HEX and better understand its application to dwelling spaces and how appropriate its use would be.

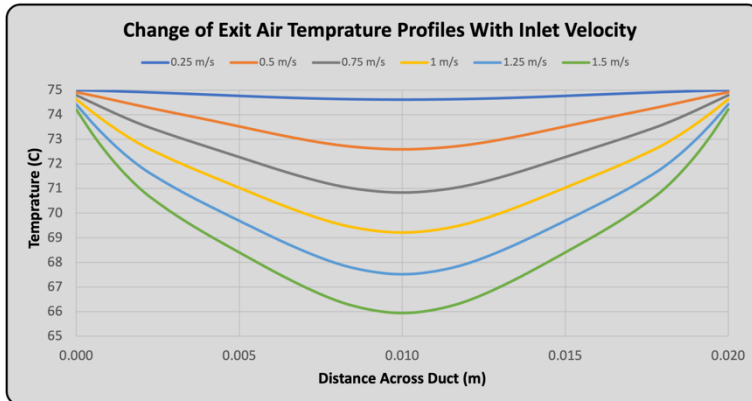


Figure 21: Temperature Profiles of Varying Velocities

becomes less constant. This effect is like what has been seen previously in the optimisation of fin spacing where an increase in fin spacing caused the temperature profile to be less constant and have a lower temperature at the centre of the duct. The $0.25 \frac{m}{s}$ case has a nearly flat temperature profile that ranges between $75^{\circ}C$ and $74.6^{\circ}C$ the air per unit volume has a high amount of thermal energy but due to the low speed, there is a small volume of that high energy air. Whereas in the $1.5 \frac{m}{s}$ case there is a 6 times greater volume of air but there will be much less thermal energy as the air is not as hot with a temperature range of $74.2^{\circ}C$ to $66^{\circ}C$.

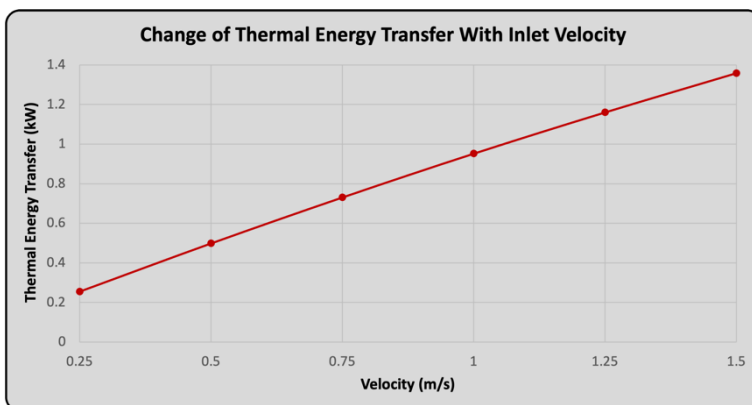


Figure 22: Thermal Energy Transfer for Different Velocities

heated allowing more energy to be delivered to the room in which the PCM is held. This gives the HEX the ability to be easily adapted to the needs of different environments by simply increasing or decreasing the airflow depending on the amount of heat needed or the period that the environment needs to be heated for. This makes the PCM HEX ideal for the dwelling space scenario because it can adapt to a magnitude of different spaces with different requirements which is very common in dwelling spaces as the amount of thermal energy that a space needs will change with the season, time of day, climate, and size of the space.

To begin, the temperature profiles of the HEX outlet have been presented in Figure 21. The profile is taken at the outlet at the centre point between fins and stretches from HEX wall to HEX wall as this method has been used consistently throughout this report and offers valuable insight. When looking at Figure 21 as the inlet velocity is increased the temperature at the centre of the duct is reduced and the overall temperature profile

To build on this a useful comparison to be made is the actual thermal energy transfer taking place. This is presented in Figure 22 where It can be seen that there is a linear relationship between the velocity of the inlet air and the thermal energy transferred to the air even though as mentioned previously the air is actually at a lower temperature in the higher velocity cases. This is down to the fact that more air is being

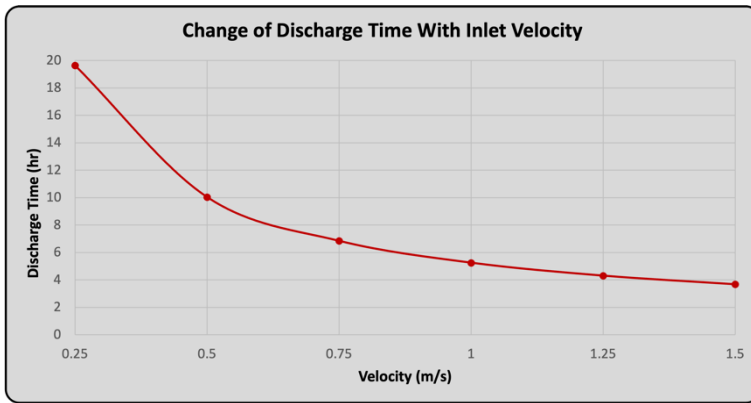


Figure 23: Discharge Time for Different Velocities

Finally, another important analysis to be made is the effect that the change in velocity will have on the discharge time of the PCM. This is presented in Figure 23 where there is a negative exponential trend so as the velocity gets higher the discharge time will get lower but with a diminishing reduction each time. The ability to discharge the HEX over different timeframes brings benefits to its application

within dwelling spaces. The HEX could be used for either heating a room quickly over a short period where the space would not need to be kept warm all day or it could be used to keep a room warm all day by discharging slowly over a bigger time frame. If the HEX is properly insulated this will allow full flexibility over the time that the PCM HEX is delivering heat to a room and allows it to be discharged when needed over a chosen timeframe.

It's important to note that the CFD analysis model used here does not incorporate the PCM/metal foam composite responsible for supplying heat to the HEX walls through congregate heat transfer. Instead, the model assigns fixed temperatures to the HEX walls as boundary conditions. This design choice ensures that the CFD model is not constrained by the rate of thermal energy supply from the PCM/metal foam composite, as would be the case in real-world conditions. The decision to omit this feature was influenced by limitations such as the one-million-node constraint in ANSYS Student and the project's time constraints. However, these omissions do not invalidate the results presented here; they still provide valuable insights into optimized duct performance and HEX behaviour under ideal conditions.

The real relationship between the Thermal transfer and the inlet velocity would be linear as seen in Figure 22 when it is within the limits of the thermal transfer of the PCM/metal foam composite but when it begins to exceed this limit the relationship would become more of a negative exponential where an increase in velocity would give diminishing returns, hence there will be a minimum time in which it is possible to discharge the PCM.

Overall, it has been found that changing the velocity of the air passing through the HEX allows for its characteristics to be changed. The HEX can be used to deliver large amounts of thermal energy over a short amount of time or smaller amounts over a large amount of time. The application of this is a domestic setting as the heating needs are rarely constant so for a HEX to be appropriate it must be able to adapt to these needs as the PCM HEX has demonstrated it can do.

Natural convection

Calculated Airflow Velocity	0.843 $\frac{m}{s}$
Average Outlet Temperature	72.01°C
Thermal Energy transfer	0.814 kW
Discharge time	6.14 hours

Table 3: Characteristics of Natural Convection Case

For this final section on the effect of velocity, the case of using natural convection only for the flow within the HEX was tested. As discussed in the method the velocity of the flow was approximated and this was used to setup an ANSYS simulation to analyse the properties of this case.

After completing the calculations, the velocity caused by natural convection was approximated to be 0.843 m/s as presented in Table 3. Due to the calculated result only approximating natural convection on a flat plate, it is reasonable to expect the actual velocity to be between this value and 0.5 m/s due to frictional losses in the duct. For this case the data will be analysed with this knowledge by using the case ran with 0.843 m/s inlet velocity and looking back at the previously completed case that ran with 0.5m/s velocity to analyse the performance of the HEX over the range of values expected under natural convection.

From this, we can understand that the discharge time of the HEX through natural convection only is between 6.14 hours to 10 hours found in the 0.843 $\frac{m}{s}$ case and the 0.5 $\frac{m}{s}$ case respectively and a thermal energy transfer rate of between 0.5kW and 0.814kW. These results of discharge times and rates of thermal energy transfer demonstrate that the PCM HEX can discharge at a reasonable rate under natural convection without the use of a fan to induce forced convection. It would also be possible to extend these times for use within dwelling spaces over a timeframe such as 12 hours by simply restricting the inlet flow to reduce the velocity of the air that travels through the HEX making them easily adjustable giving flexible application.

The application of the HEX without a fan would make them more suitable for use within dwelling spaces as the overall mass would be reduced greatly when combined with the reduction of the number of fins from the optimisation previously performed. On top of that the production cost and energy usage to make them a more affordable solution to thermal energy storage. However, a smaller fan could still be included in the HEX in order to increase the rate of thermal discharge over the HEX. This makes the PCM HEX an extremely adaptable heat storage unit that can hold thermal energy efficiently and deliver it over a range of different periods.

6. CONCLUSIONS

Through the CFD analysis of the effectiveness of the inclusion of fins, it was found that the fins performed exceptionally well in enhancing the rate of heat transfer to the air. It was noted that with the original fin spacing, all the heat transfer takes place within the first half of the duct.

From this, an analysis was performed to find the optimal fin spacing considering both thermal behaviour and pressure drop. It was found that an 8mm spacing is optimal as this allowed the air to gain thermal energy along the entire duct and caused a reduction in pressure drop by a factor of a half compared to the previous fin spacing of 5mm. This increase in fin spacing allows a reduction in pressure drops which will help to reduce the overall mass and cost of the HEX as a lighter, less expensive fan can be used. To ensure accuracy, mathematical validation was performed on the CFD models used in the Optimisation of the fins work package using the one-dimensional heat transfer equation and fin theory.

Finally, the effect of velocity was assessed along with a case assessing the HEX using only natural convection. The main conclusion was that the HEX performs effectively over a range of velocities, this can be utilised to alter the discharge rate of the PCM making the HEX adaptable for use within dwelling spaces where the heating needs are inconsistent. It was also found in the natural convection case that the HEX can be run without a fan by using natural convection alone whilst maintaining a discharge time of 6 to 10 hours. The combination of these conclusions and the previous optimisation findings allows the HEX to be more suited to dwelling spaces as its mass and size can be reduced, along with production costs making it an effective solution to thermal energy storage.

7. FUTURE WORK

After completing the effect of velocity work package there is one clear area for future work to take place. Previously, the natural convection in the duct was found using order of magnitude approximations that are based on a flat plate, not a duct, this means that the accuracy is limited. To build on this and reach more accurate conclusions a good next step would be to construct a complete CFD model that allows for the simulation of natural convection. This will require a much larger CFD model that is likely to go over the one million node limit for ANSYS student and it is likely to be much more complex to set up reliably compared to the simulations already conducted within this project.

Doing this would give a far more detailed and reliable analysis of the behaviour that the PCM HEX exhibits when natural convection is the only force drawing air through the duct. This would in turn give stronger indications of the suitability of removing the fan on the HEX within dwelling space applications.

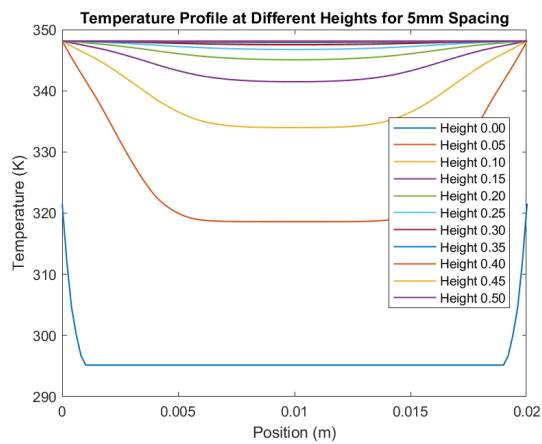
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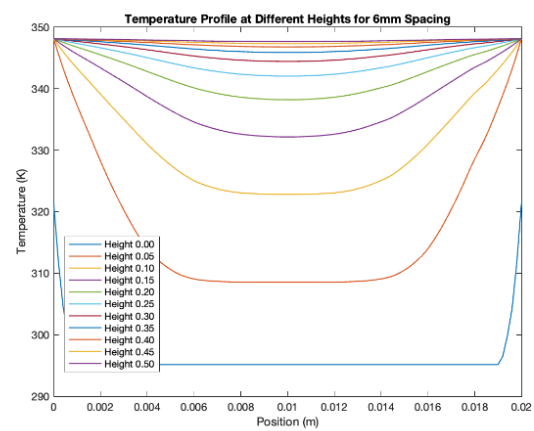
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9. APPENDIX

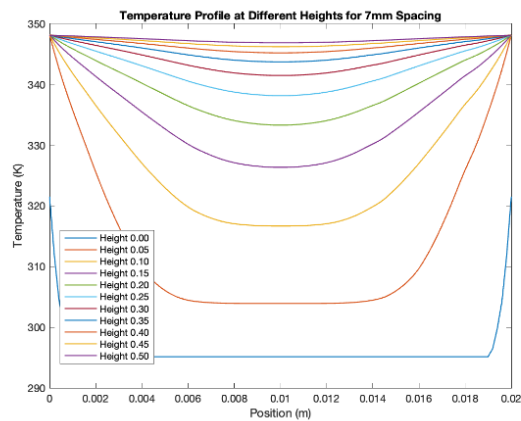
Optimisation of Fins – Refined Test Temperature Profiles



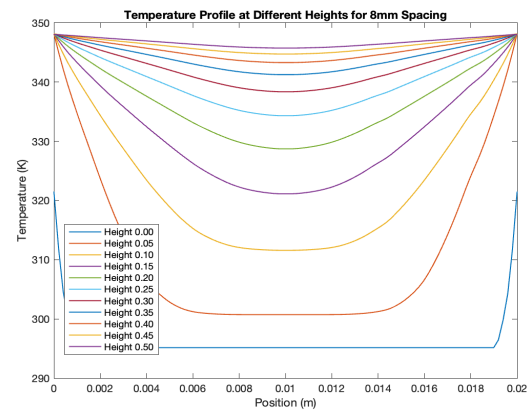
Temperature Profile for 5mm Fin Spacing



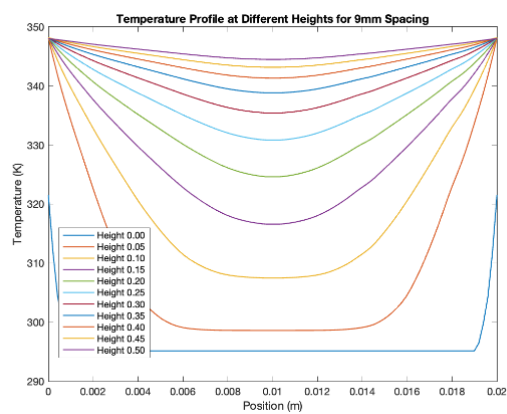
Temperature Profile for 6mm Fin Spacing



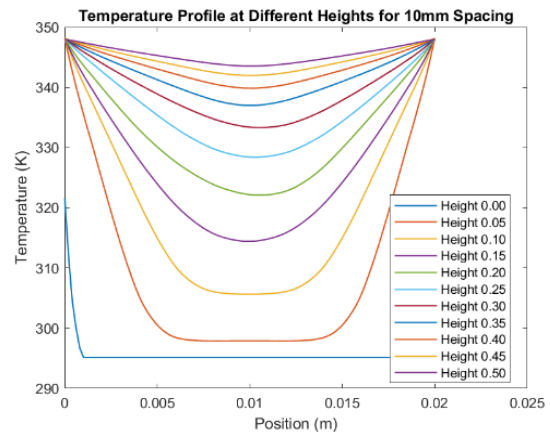
Temperature Profile for 7mm Fin Spacing



Temperature Profile for 8mm Fin Spacing



Temperature Profile for 9mm Fin Spacing



Temperature Profile for 10mm Fin Spacing